**DELHI TECHNOLOGICAL UNIVERSITY**
**(Formerly Delhi College of Engineering)**
**Shahbad Daulatpur, Bawana Road, New Delhi 110042)**

# DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

# B.TECH - PROJECT
# DETAILED RESEARCH PLAN

**Detailed Research Plan**

**Can an AI Explanation Be Misused? Testing Whether SHAP and LIME Explanations Help Someone Cheat a Clinical Risk Model**

## 1. Title of Project

Can an AI Explanation Be Misused? Testing Whether SHAP and LIME Explanations Help Someone Cheat a Clinical Risk Model

This continues the approved synopsis, "Evaluating the Trustworthiness of Explainable AI Methods in Clinical Risk Prediction". The synopsis listed adversarial robustness of explanations as planned future work; this plan takes that up as the main study.

## 2. Group Id and Team Details

| **Group Id** | **GP-109** |
|---|---|

### 2.1 Details of all the Members (Final Year, B.Tech CSE)

| **S. No.** | **Name** | **Roll Number** |
|---|---|---|
| 1 | Akash Singh | 23/CS/036 |
| 2 | Aditya Jha | 23/CS/025 |
| 3 | Akshat Kumar | 23/CS/039 |

### 2.2 Name of the Guide and Co-guide

| **Name of the Guide** | Miss. Anjali Bansal, Department of Computer Science & Engineering |
|---|---|
| **Name of the Co-guide** | - |

### 2.3 Junior Team Members (from 2nd / 3rd year, minimum 1 required)

| **S. No.** | **Name** | **Roll Number** | **Year** |
|---|---|---|---|
| 1 | Ishan Singh | 25/CS/168 | 2nd Year |

## 3. Abstract / Brief Description / Problem Statement

Hospitals and insurers increasingly use machine learning to score a patient's health risk, and laws often require the system to explain each score, usually with SHAP or LIME, which show how much each input contributed to the result. Some researchers worry that an explanation also works as instructions, telling a person which answers to change to get a better score, and so suggest hiding it; but a person can also just try a few different answers and watch the score move, and nobody has tested whether the explanation helps any more than that. We compare two situations on the same patients: one person is shown the explanation, the other sees no explanation but gets the same number of attempts at the model, and we report how many ordinary attempts the explanation is worth. We use NHANES, a public US health survey that records height and weight both as self-reported and as measured by a nurse while the diabetes result comes from a blood test, so we train two models identical except for this one column and use the real reported-vs-measured gap to decide how large a realistic lie is. If explanations add little, they can be shown openly and checking inputs is the better safeguard; if they add a lot,

our comparison shows which data is responsible. We will publish our plan before the final experiment so the conclusion cannot be adjusted afterwards.

## 4. Brief Literature Review

LIME (Ribeiro et al., KDD 2016) and SHAP (Lundberg & Lee, NeurIPS 2017) are the two explanation methods we study; one common version of SHAP is mathematically very close to LIME, so we treat them as a pair. Explanations can look convincing yet be disconnected from the model (Adebayo et al., NeurIPS 2018), and Bilodeau et al. [1] proved that for certain models they can do no better than random guessing, which shows a problem is possible but not how large it is in practice. Milli et al. (FAT* 2019) showed that ten explanation queries can match a thousand ordinary ones, and Shokri et al. [4] found similar privacy risks, but both protect the model or training data rather than address a patient changing their own answers. Ghalme et al. [5] found that hiding the decision rule usually does not help the decision-maker, and Hikri & Rosenfeld [2] extended this to partial disclosure forced by regulation, which suggests transparency is usually better. König et al. [3] noted that model advice can improve a prediction without making someone healthier, just as lying about weight is not losing weight, and Bordt et al. (FAccT 2022) argue that explanations suit legal transparency rules poorly because providers can manipulate them, so we only measure disclosure and do not propose weakening it. No study yet measures how much an explanation helps a person improve their own score compared with the same number of ordinary attempts, and that is the gap this project fills.

## 5. References

Five references from reputed conferences and journals, all from the last five years. Older foundational papers mentioned in Section 4 (Ribeiro 2016, Lundberg & Lee 2017, Adebayo 2018, Milli 2019) are not counted here.


**[1]** B. Bilodeau, N. Jaques, P. W. Koh, and B. Kim, “Impossibility theorems for feature attribution,” Proceedings of the National Academy of Sciences (PNAS), vol. 121, no. 2, e2304406120, 2024.

**[2]** I. Hikri and N. Rosenfeld, “Ambiguous Strategic Classification,” in Proceedings of the 43rd International Conference on Machine Learning (ICML), 2026.

**[3]** G. König, T. Freiesleben, and M. Grosse-Wentrup, “Improvement-Focused Causal Recourse (ICR),” in Proceedings of the AAAI Conference on Artificial Intelligence, vol. 37, no. 10, pp. 11847–11855, 2023.

**[4]** R. Shokri, M. Strobel, and Y. Zick, “On the Privacy Risks of Model Explanations,” in Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society (AIES), pp. 231–241, 2021.

**[5]** G. Ghalme, V. Nair, I. Eilat, I. Talgam-Cohen, and N. Rosenfeld, “Strategic Classification in the Dark,” in Proceedings of the 38th International Conference on Machine Learning (ICML), PMLR 139, pp. 3672–3681, 2021.


## 6. Technical Requirements

### 6.1 Hardware

| Item | Requirement |
|---|---|
| **Computers** | Three personal laptops (8–16 GB RAM). No special hardware needed. |
| **GPU** | Not required. We use standard machine learning models on tabular data, not deep learning. |
| **Storage** | Under 5 GB in total. |

### 6.2 Software

| Purpose | Tool |
|---|---|
| **Programming language** | Python 3.9 or later |
| **Data handling** | pandas, numpy |
| **Machine learning models** | scikit-learn (logistic regression, random forest) |
| **Explanations** | shap, lime |
| **Statistics** | scipy, statsmodels |
| **Graphs and charts** | matplotlib, seaborn |
| **Code management** | Git, with fixed random seeds so results can be reproduced |

### 6.3 Computing Time

| Task | Approximate time |
|---|---|
| **Preparing data and training models** | A few minutes. |
| **Generating explanations** | A few minutes, using the fast version of SHAP for tree models. |
| **Main experiments** | About 6–11 hours of processing, which can be left running overnight. |
| **Total for the project** | Roughly 30–50 hours of computer time, all on our own laptops. |

### 6.4 Data and Ethics

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|---|---|
| **Main dataset** | NHANES (US Centre for Disease Control), survey rounds 2015–2016 and 2017–2018. Free, public, and anonymous. |
| **Supporting datasets** | CDC Diabetes Health Indicators (UCI dataset 891) for comparison, and UCI Heart Disease (dataset 45) for testing our code. |
| **Ethics approval** | Not required. All data is already public and anonymous. We do not collect data from any person, and we make no medical recommendations. |

## 7. Expected Outcome with Timeline / Work Plan

### 7.1 What We Expect to Produce

| Output | Description |
|---|---|
| **Working software** | A Python pipeline that trains the risk models, produces SHAP and LIME explanations, and measures how much an explanation helps someone change their score. |

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|---|---|
| **Prepared datasets** | The NHANES survey and examination files combined, with the self-reported and measured columns documented. |
| **Reported vs measured study** | A measurement of how much people's self-reported height and weight differ from the nurse's measurement, used to decide how large a realistic lie is. |
| **Published analysis plan** | Our method and expected result posted publicly and time-stamped before the final experiment, so the conclusion cannot be adjusted afterwards. |
| **Main result** | A number showing how many ordinary attempts an explanation is worth, with clear graphs. |
| **Comparison of the two models** | A direct measurement of how much difference it makes when an input is self-reported instead of measured. |
| **Project report** | Final year project thesis and viva presentation. |

### 7.2 What We Expect to Find

We expect to find that explanations give only a small advantage compared with simply trying the model a few times. If that is the result, it suggests that hiding explanations is not an effective safeguard, and that checking patients' reported information is more useful. If instead we find a large advantage, our two-model comparison will show which part of the data causes it. Both results are worth reporting, and we will state our expectation publicly before running the final experiment.

### 7.3 Work Plan

| Phase | Period | Work |
|---|---|---|
| **P1** | 8 Oct – 21 Oct 2026 | Set up the environment and test the code on a small dataset where results can be checked by hand. |
| **P2** | 15 Oct – 14 Nov 2026 | Study the NHANES documentation, decide which columns to use, and mark which ones are self-reported. |
| **P3** | 1 Nov – 30 Nov 2026 | Prepare the data: combine the survey and examination files, and measure how much reported values differ from measured ones. |
| **P4** | 8 Nov – 7 Dec 2026 | Build the main experiment code and train the two models. |
| **P5** | 1 Dec – 14 Dec 2026 | Trial run on the 2015–2016 data. Produce the first results and graphs. |
| **P6** | 1 Jan – 14 Jan 2027 | Finalise the experiment settings from the trial run and publish the analysis plan online with a time stamp. |
| **P7** | 8 Jan – 7 Mar 2027 | Run the final experiments on the 2017–2018 data, which has not been examined before this point. |
| **P8** | 1 Mar – 31 Mar 2027 | Analyse results, prepare graphs and tables, and check every reference against the original paper. |
| **P9** | 15 Mar – 30 Apr 2027 | Write the research paper and submit it to the chosen conference. |

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|---|---|---|
| **P10** | 15 Apr – 31 May 2027 | Write the thesis and final project report. |
| **P11** | 22 May – 14 Jun 2027 | Prepare for the viva and make the final submission. |

## 7.4 Gantt Chart

| Activity | Oct 26 | | Nov 26 | | Dec 26 | | Jan 27 | | Feb 27 | | Mar 27 | | Apr 27 | | May 27 | | Jun 27 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 2 |
| P1 Setup and code testing | ■ | | | | | | | | | | | | | | | | | |
| P2 Study data documentation | | ■ | ■ | | | | | | | | | | | | | | | |
| P3 Prepare the datasets | | | ■ | ■ | | | | | | | | | | | | | | |
| P4 Build experiment code | | | ■ | ■ | ■ | | | | | | | | | | | | | |
| P5 Trial run and first results | | | | | ■ | | | | | | | | | | | | | |
| P6 Publish analysis plan | | | | | | | ■ | | | | | | | | | | | |
| P7 Final experiments | | | | | | | ■ | ■ | ■ | ■ | ■ | | | | | | | |
| P8 Analysis and graphs | | | | | | | | | | | ■ | ■ | | | | | | |
| P9 Write and submit paper | | | | | | | | | | | | ■ | ■ | ■ | | | | |
| P10 Write thesis | | | | | | | | | | | | | | ■ | ■ | ■ | | |
| P11 Viva preparation | | | | | | | | | | | | | | | | ■ | ■ | |
| *Spare time / buffer* | | | | | | ▒ | | | | | | | | | | | | ▒ |

Black = work in progress. Grey = spare time kept aside for delays.